Friedrich-Alexander-Universität
Technische Fakultät

Stress-testing Deep Learning Models on Sparse Laboratory Data

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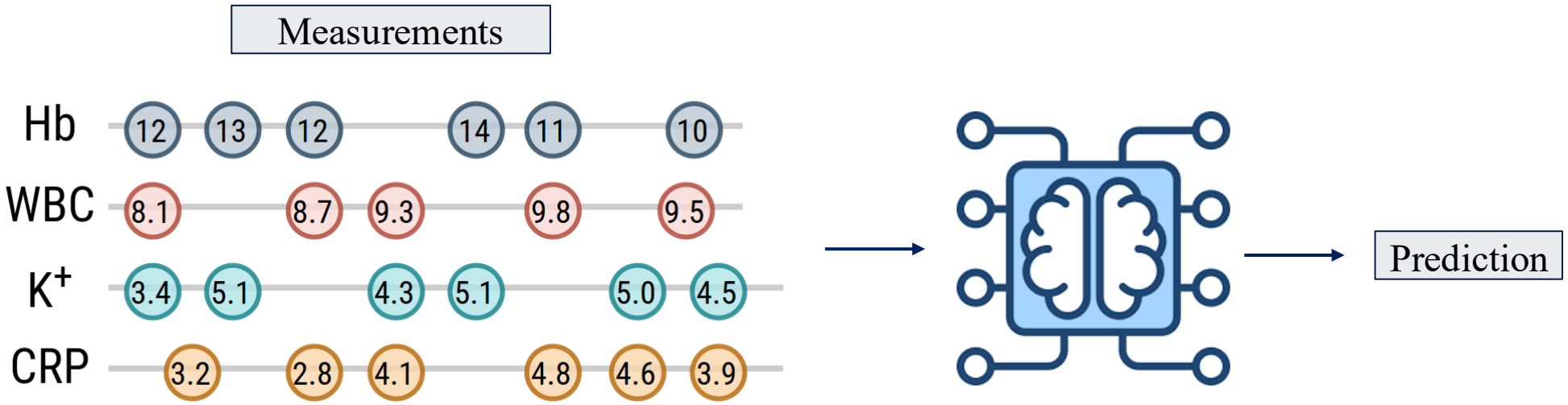
Laboratory Data

What is Laboratory Data?

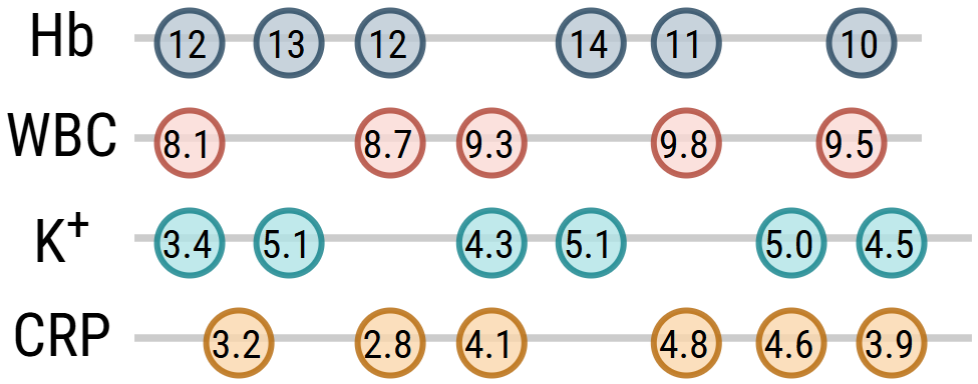
- Numerical results from medical tests (e.g. blood, urine, or tissue analysis)
- Traditionally used by clinicians to make diagnoses
- Recently used to train ML models to make predictions
- Common measurements: Hemoglobin (Hb), White Blood Cells (WBC), Potassium (K^+), C-Reactive Protein (CRP)



Laboratory Test Frequency

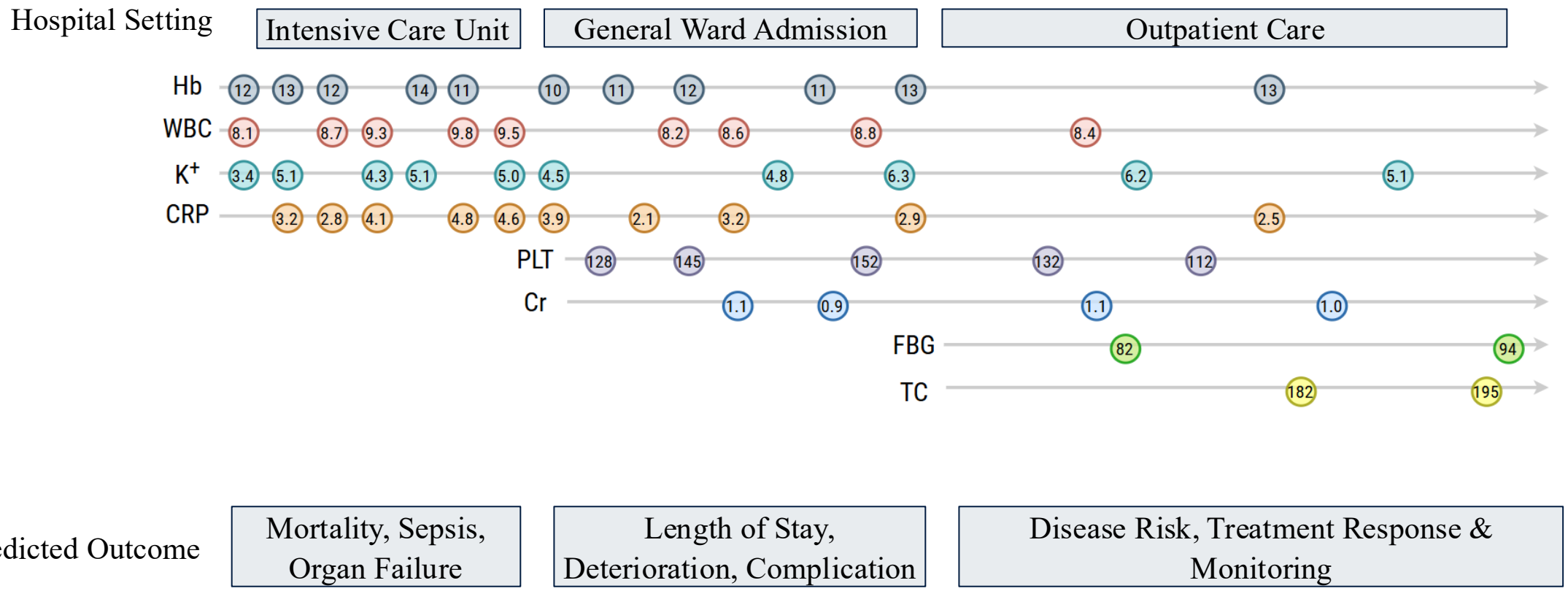
Type of Test

- Multiple times a day → K^+, Blood glucose, Blood gases (ABG)
- Once per day → Hb (Hemoglobin), WBC (White Blood Cells), Sodium (Na^+)
- Once per week → Vitamin B12, Liver function tests (ALT, AST)
- Once per month → HbA1c, Cholesterol panel, Thyroid hormones (TSH, T3, T4)



Laboratory Test Frequency

Type of Care Settings



Intensive Care Unit (ICU) Cohorts

What is the ICU?

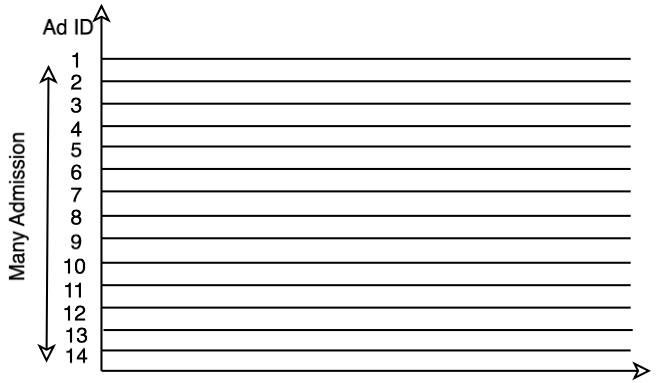
- Patients who are critically ill and need constant monitoring
- Serious infections, organ failure, severe injuries, heart attacks
- ICU Cohorts typically include all ICU patients

What is ICU data?

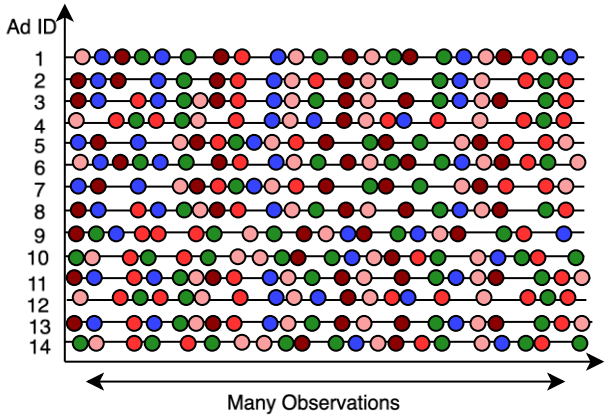
Vital Signs (monitored continuously): HR, BP, RR, Temperature

Laboratory measurement (frequently): ABG, K^+, Na^+, Ca^{2+}, ALT, AST

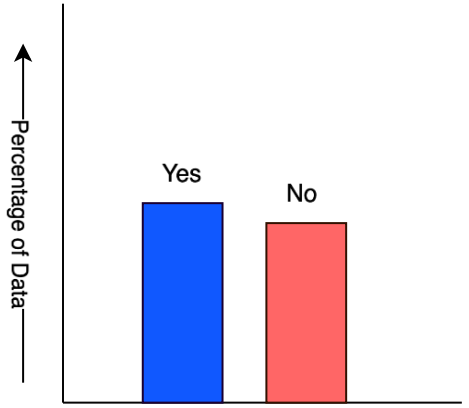
Rich in Observations



Densely Sampled



Balanced in Target



Non-ICU Cohorts

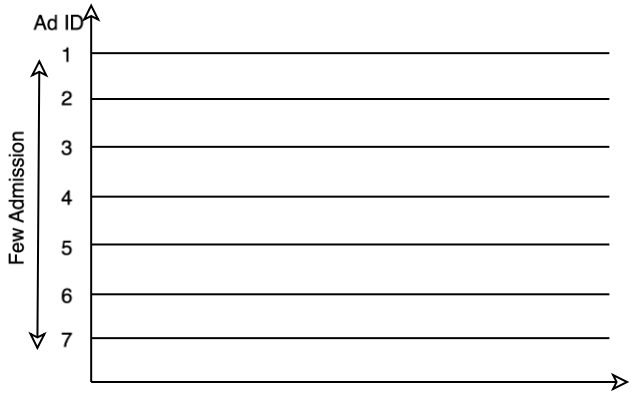
What is Non-ICU data?

- Not in the Intensive Care Unit
- Taken from general wards, outpatient clinics, or scheduled treatments
- Non-ICU cohorts typically include a small subset of admissions (patients with cancer diagnosis who receive chemotherapy)

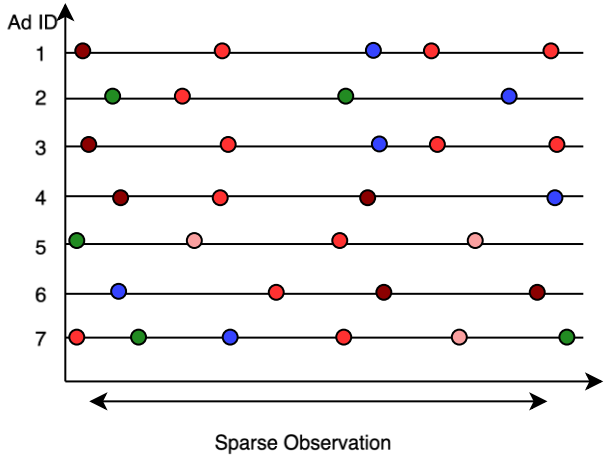
Types of data:

- Laboratory measurement (less frequent)

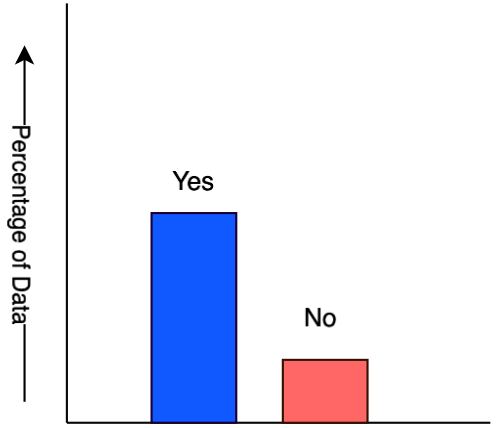
Sparse in Observations



Irregularly Sampled



Imbalanced in Target



Motivation

- ✓ Most studies have used ICU data to develop and evaluate models for prediction tasks, a relevant example is STraTS (Tipirneni et al., 2022) [1]
- ✓ A recent study by Rahimi et al. (2025) [2] tested these models on two non-ICU cohorts for the prediction of chemotherapy side effects (aplasia, Neutropenic fever) and showed lower performance, especially in the neutropenic fever cohort
- ✓ Results suggest that these models perform significantly worse when applied to non-ICU data

Motivation

- To analyze how models behave in non-ICU settings
- To assess which perturbation—smaller sample size, sparsification, or class imbalance—leads to the greatest performance drop
- To understand which model should be preferred for each data scenario

[1] S. Tipirneni and C. K. Reddy, "STraTS : Self-supervised transformer for time-series representation learning in healthcare," in *Proc. AAAI Conf. Artif. Intell.*, 2022.
[2] Rahimi *et al.*, "Non-temporal tree-based models outperform temporal deep learning models in the prediction of chemotherapy-induced side effects from longitudinal laboratory data," 2025.

Methodology

Datasets:

Statistic	MIMIC-III	Physionet-2012
Number of ICU stays	52,871	11,988
Temporal Variables	129	37
Demographics	Age, Gender	Age, Gender, Height, ICU Type
Task	24-hour mortality	48-hour mortality
Target	In-Hospital Mortality with 9.7% positive class, Length of Stay with avg of 4 days	In-Hospital Mortality with 14.2% positive class, Length of Stay with avg of 13 days

Train/Validation/Test split→ (Tipirneni et al., 2022)

S. Tipirneni and C. K. Reddy, "STraTS: Self-supervised transformer for time-series representation learning in healthcare," in *Proc. AAAI Conf. Artif. Intell.*, 2022.

Methodology

Models:

Model	Full Model Name	Type	Input
GRU	Gated Recurrent Unit	RNN	Discretized regular time series (mean value, missing mask, time since last observed)
GRU-D	GRU with Decay	RNN	Irregular time series (measured value, missing mask, delta time)
TCN	Temporal Convolutional Network	CNN	Discretized regular time series (mean value, missing mask, time since last observed)
SAnD	Simply Attend and Diagnose	Transformer	Discretized regular time series (mean value, missing mask, time since last observed)
STraTS	STraTS (Sparse time-series transformer model)	Transformer	Measurement events (triplet value variable timestamp)

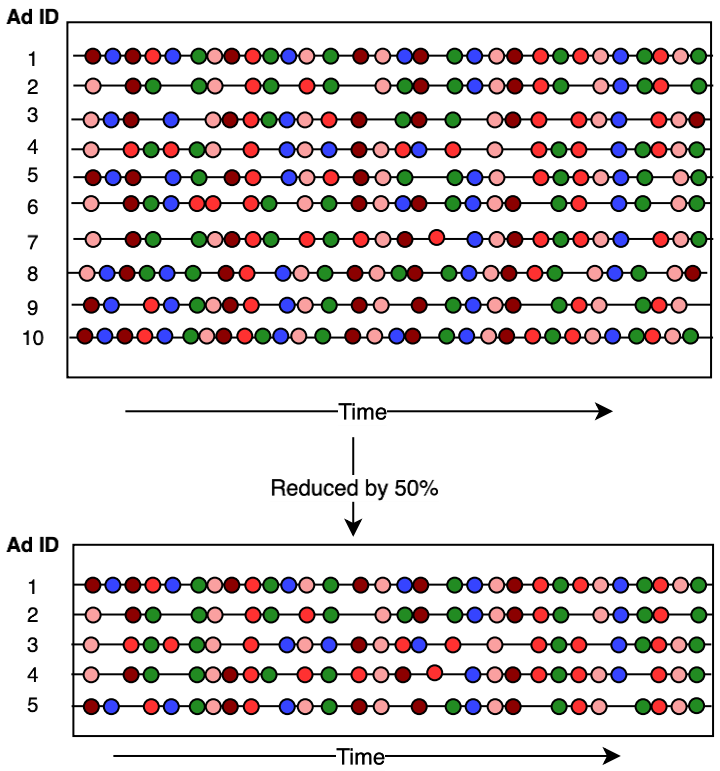
Excluded : InterpNet → slow, poor performance in Rahimi et al. (2025), unable to interpolate sparse data

Rahimi *et al.*, "Non-temporal tree-based models outperform temporal deep learning models in the prediction of chemotherapy-induced side effects from longitudinal laboratory data," 2025

Perturbation Approach

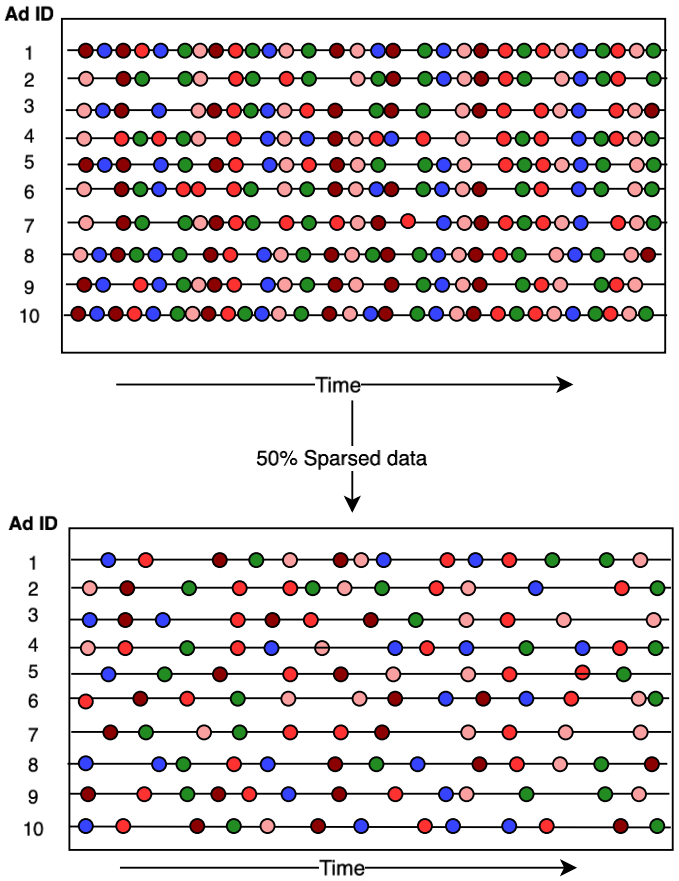
Subsampled:

Reduces the dataset by selecting a subset of data



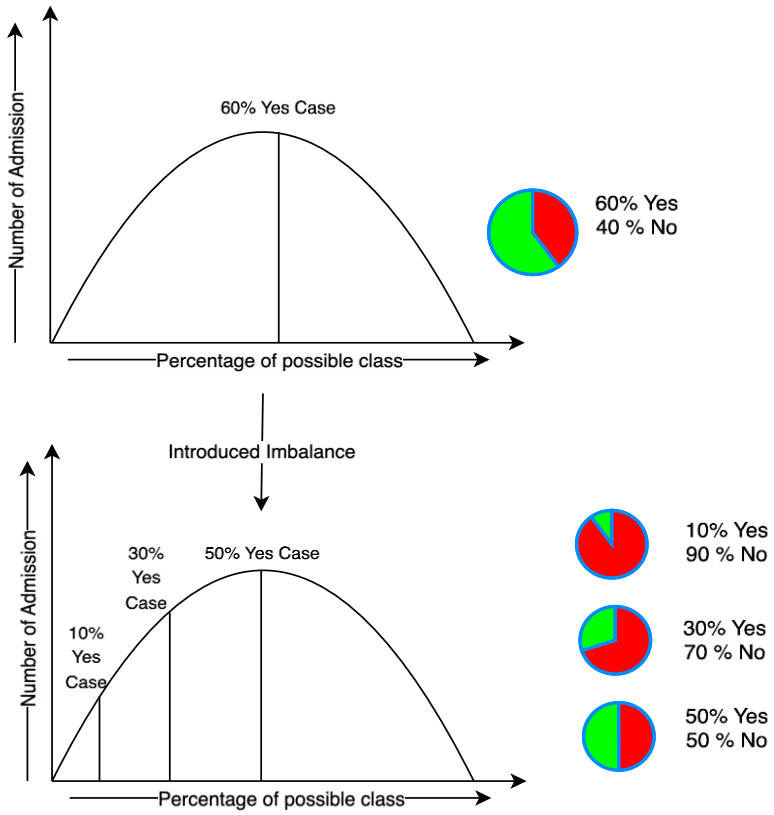
Sparsification:

Randomly removes measurements



Imbalanced:

Imbalances the dataset by increasing the instances of one class



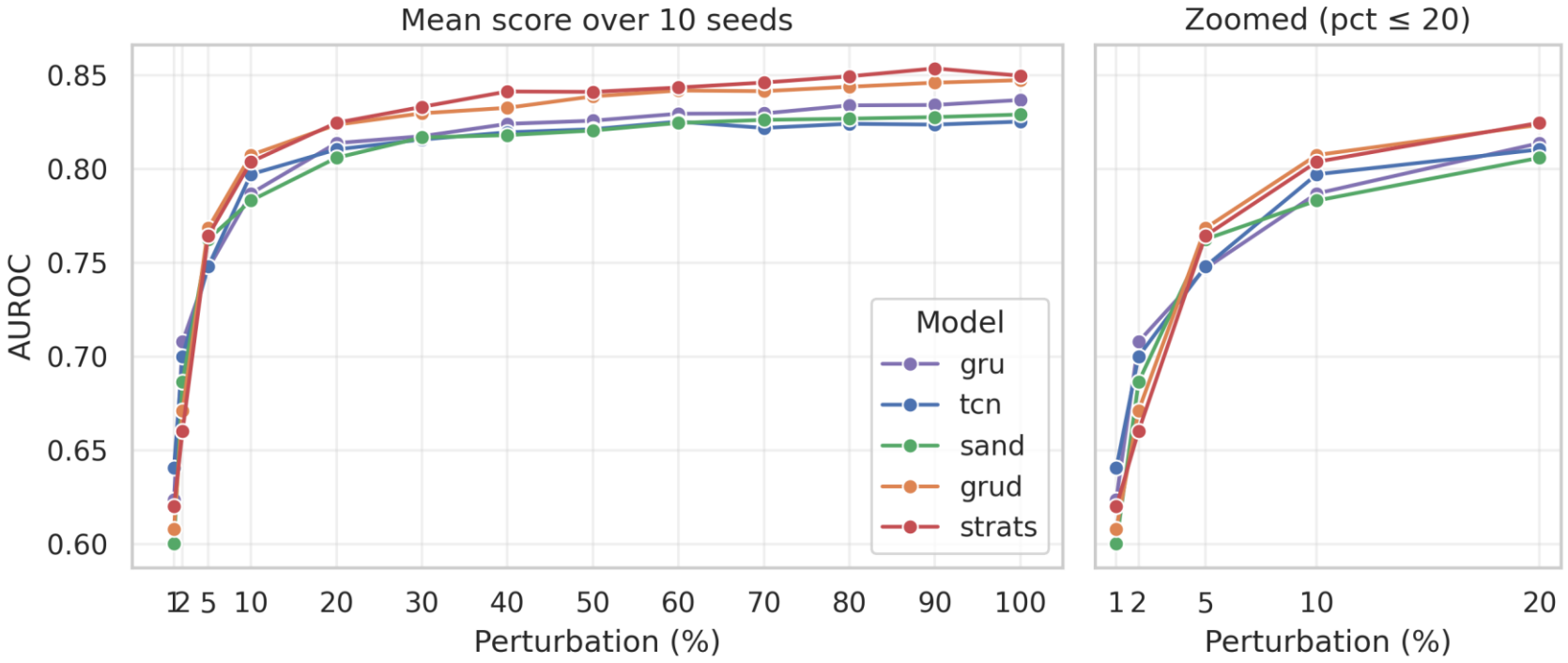
Perturbation Approach

Experimental Setup

Perturbation Details	Subsampled (reduced cohort size)	Sparsification (increased observation sparsity)	Imbalanced (label imbalance via LOS thresholding)
Description	Reduces the size of training and validation cohorts by selecting a random subset of admission IDs.	Increases the sparsity of time-series data by removing measurements while keeping demographics unchanged.	Creates label imbalance by converting the continuous Length-of-Stay (LOS) value into a binary target
Target	In-Hospital-Mortality	In-Hospital-Mortality	Length-of-Stay (LOS)
Percentage	$p \in \{1,2,5,10, 20, 30,40,50,60,70,80,90,100\}$	$p \in \{1,2,5,10, 20, 30,40,50,60,70,80,90,100\}$	$p \in \{1,2,3,4,5,6,7,8,9,10, 20, 30,40, 50\}$
Seed	10 random seed	10 random seed	No seed
Key Process	**Randomly select p% of admission IDs** from training and validation, keep test set fixed.	**Retain p% of measurements** for each admission-variable pair	**Calculate a threshold T based on the desired positive-class rate p** for the training split. Apply this threshold to the validation and test splits

Results

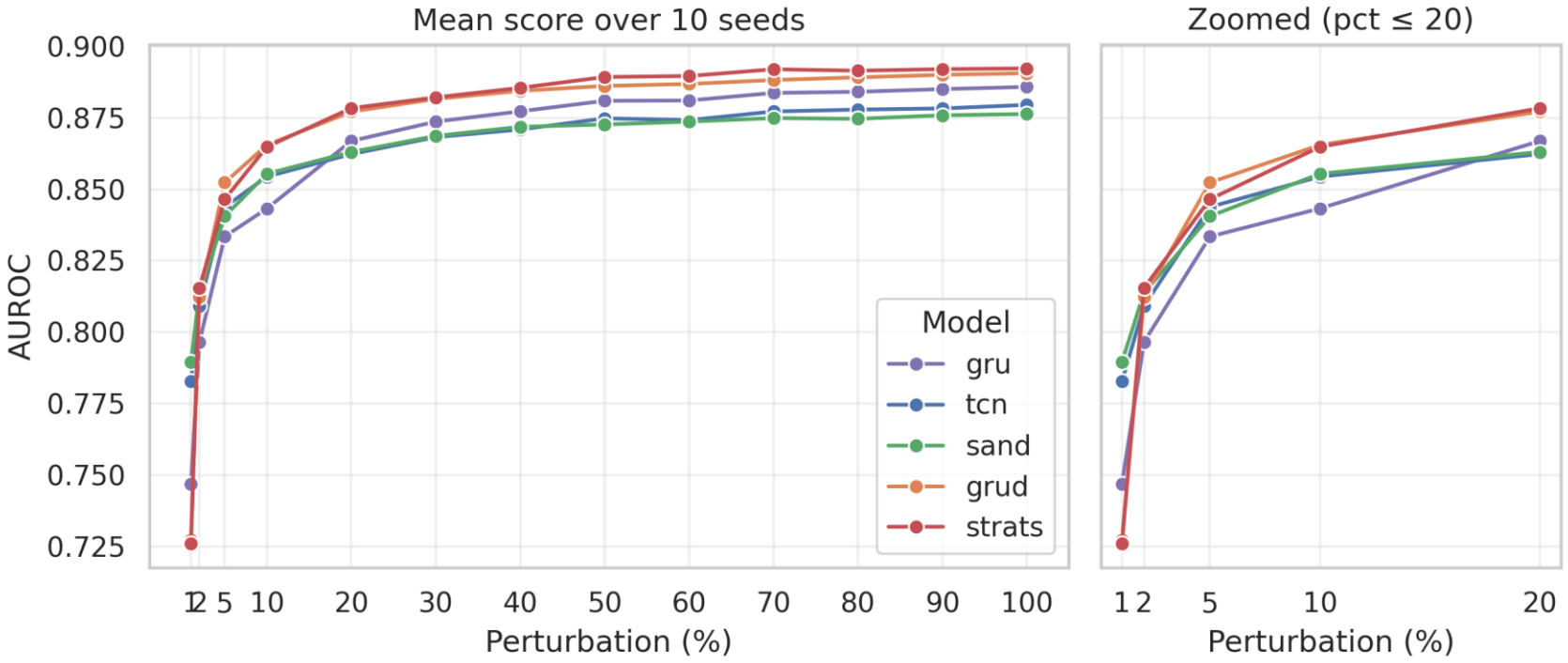
Subsampled (Physionet-2012)



- Model performance decreases below 20% of the data (~2000 admissions) and drops below 10%
- GRU-D and STraTS perform considerably worse with only 5% of the data

Results

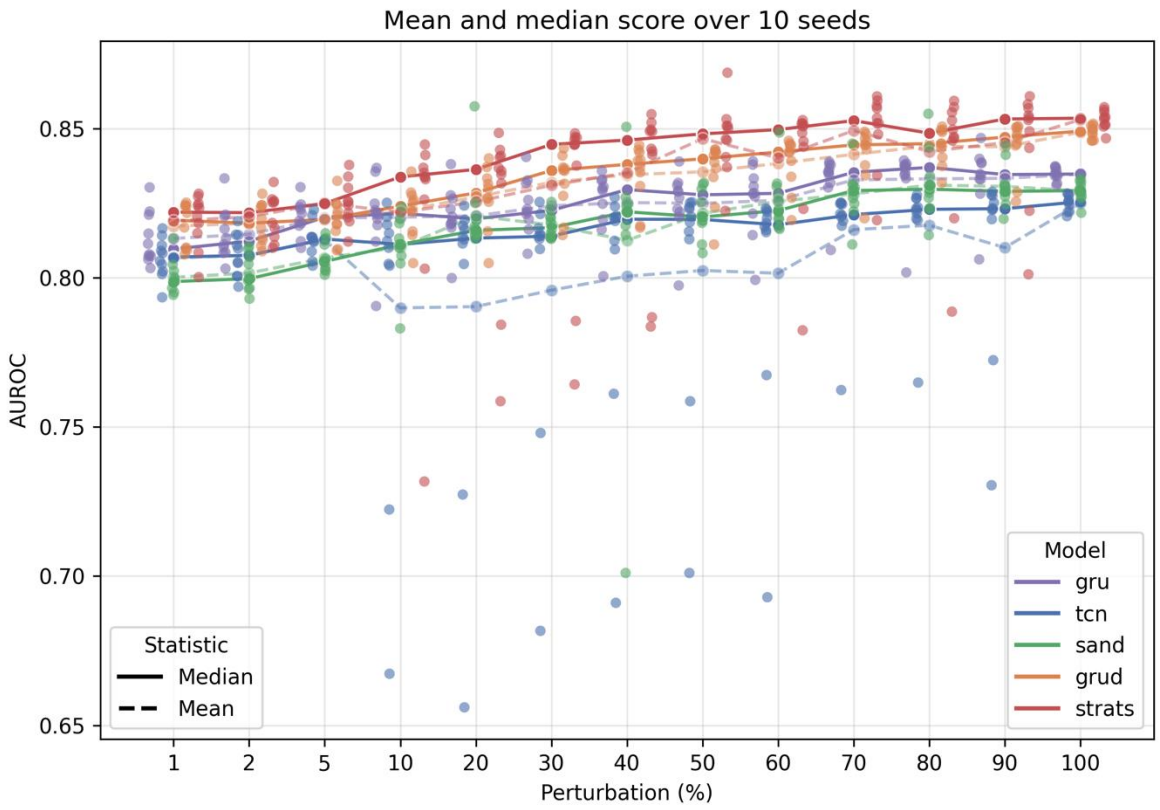
Subsampled (Mimic-III)



- Model performance decreases between 10% and 20% of the data (~5000 admissions) and drops below 5%
- GRU-D and STraTS perform considerably worse with only 1% of the data

Results

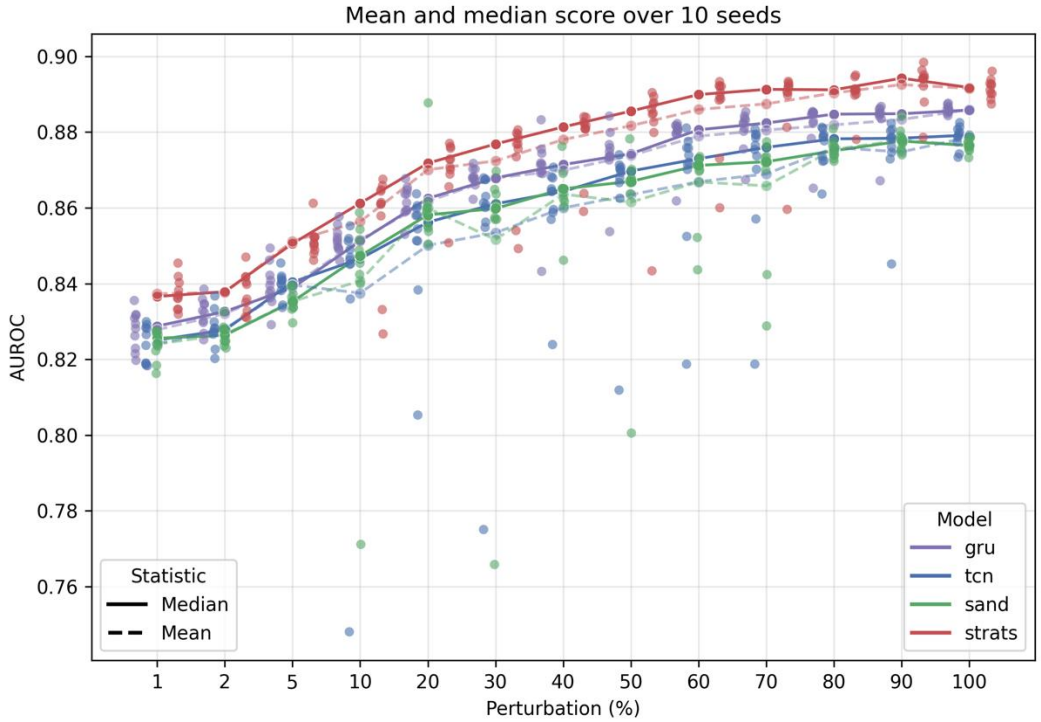
Sparsification (Physionet-2012)



- Performance decreases slightly but steadily with increased sparsification
- Performance drops for certain seeds (outliers)

Results

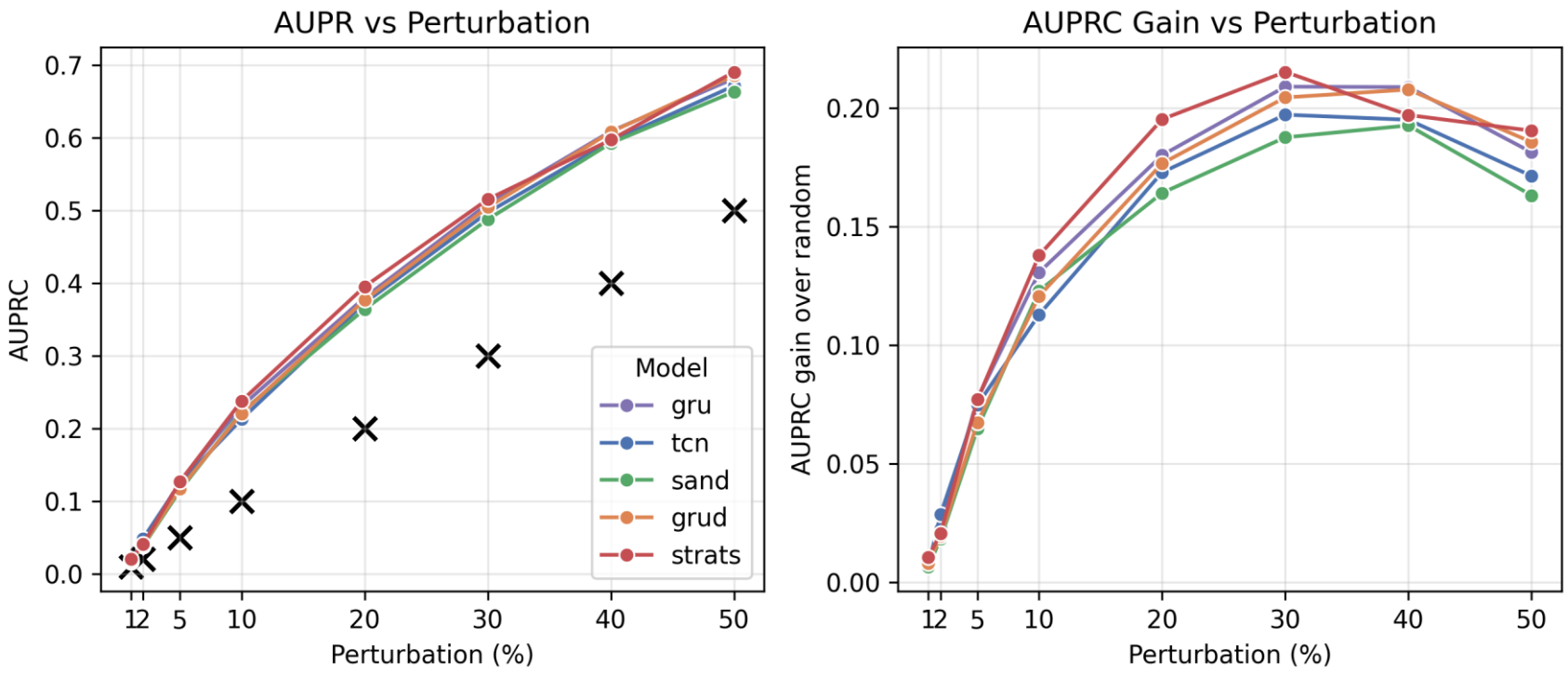
Sparsification (Mimic-III)



- Performance decreases slightly but steadily with increased sparsification
- Performance drops for certain seeds (outliers)
- GRU-D breaks down under sparsification, as it requires at least two measurements to compute delta values

Results

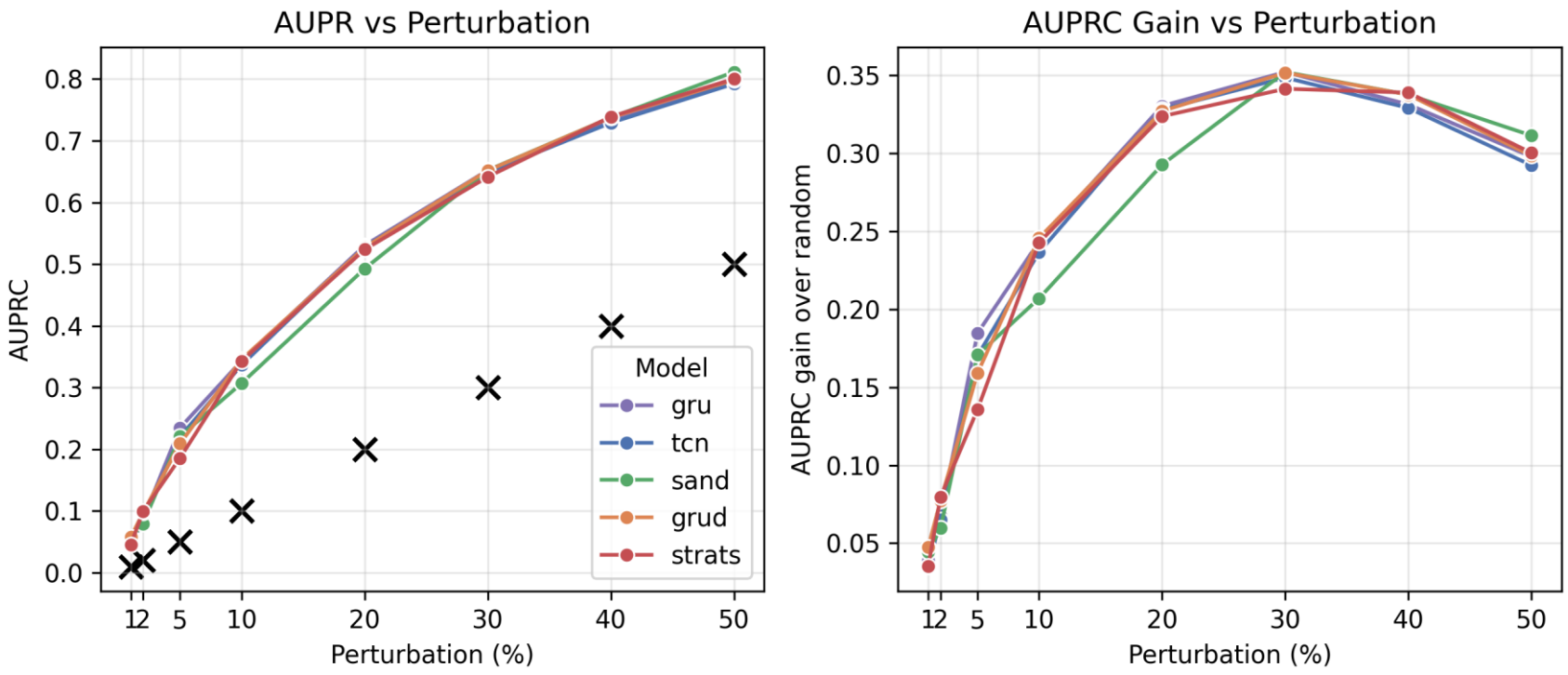
Imbalanced (Physionet-2012)



- Performance decreases below 30% positive class and drops below 20%

Results

Imbalanced (Mimic-III)



- Performance decreases below 30% positive class and drops below 20%

Results

Comparison with NF cohort

	MIMIC IV NEUTROPENIC FEVER (NF)	PHYSIONET ORIGINAL	PHYSIONET MULTIPERTURBED (perturbation applied to train and val splits only, test unchanged)
Sample Size (# admissions)	Train: 3389 Val: 878 Test: 1041 Total: 5308	Train: 7672 Val: 1918 Test: 2398 Total: 11998	Train: 3389 Val: 878 Test: 2398 (unchanged)
Measurement Sparsity (avg measurements per adm-variable)	6.2	14.5	6.5
Class Imbalance (% positive class)	4.8%	14.2%	4.8%

Training protocol

- All models are trained using **5-fold cross-validation**
- Best hyperparameters and training settings follow **Rahimi et al. (NF cohort)**

Per-fold procedure

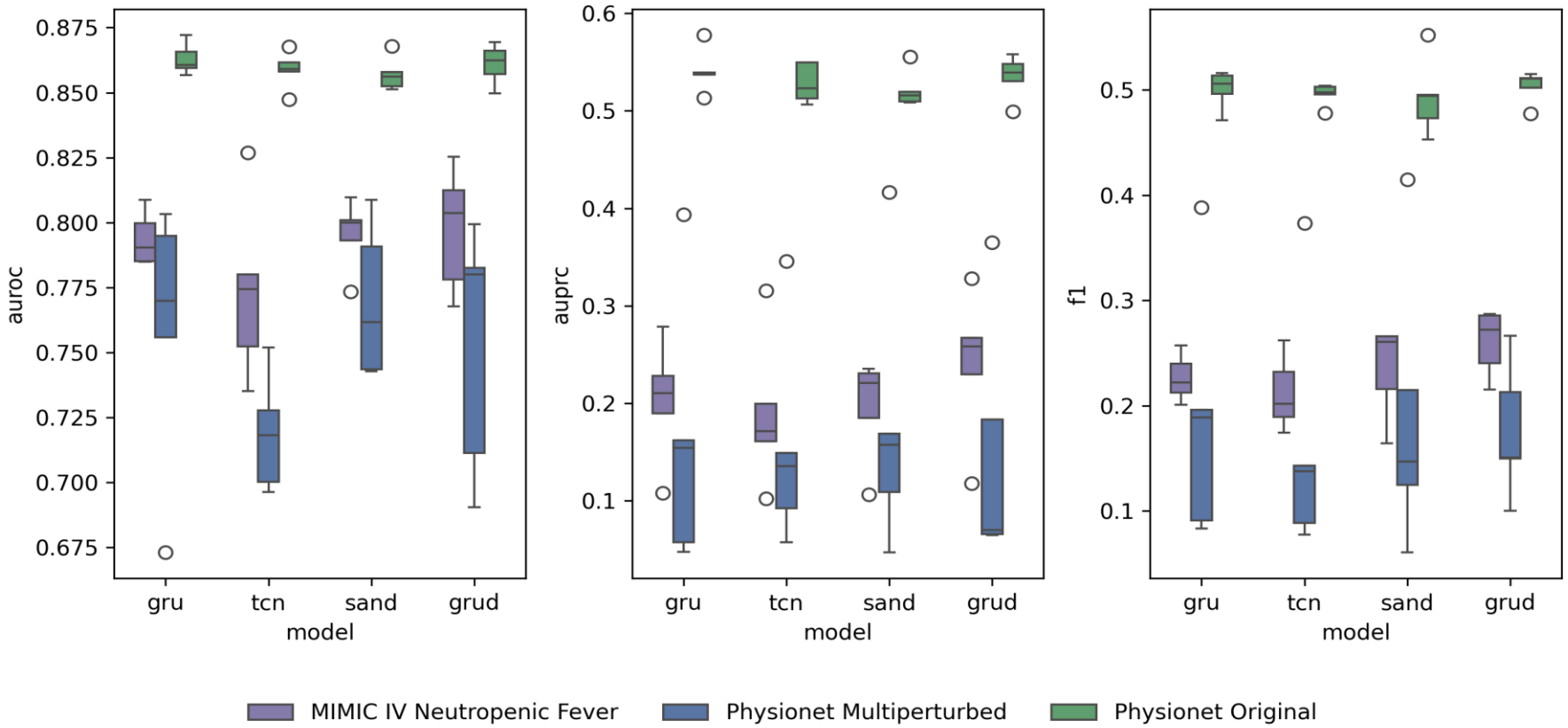
- Train on **train/validation/test split** to determine optimal epochs via **early stopping on validation set**
- Retrain on **train + validation** using selected number of epochs

Final evaluation on **test set**

Rahimi *et al.*, "Non-temporal tree-based models outperform temporal deep learning models in the prediction of chemotherapy-induced side effects from longitudinal laboratory data," 2025

Results

Comparison with NF cohort



- When data characteristics in Physionet match NF, performance drops
- This suggests that performance in NF is low not because of the task difficulty, but because DL struggle with these data characteristics

Conclusion

Conclusions:

- Model performance declines as cohort size decreases and drops at around 2000 samples
- Performance declines as the percentage of the positive class decreases and drops below 20%
- Compared with other perturbations, sparsification has a smaller impact on model performance (but there can be outliers)
- Results are comparable across models

Limitations:

- Low number of seeds because of high computational cost
- Cross-validation was applied to limited test cases
- Best parameters were taken from the paper (Tipirneni et al.), because of computational cost no hyperparameter tuning
- While other methods work on static data, GRU-D fails to compute delta intervals

Future Work:

- Subsampling both patient IDs and variables to assess the impact on model performance
- Systematic combined perturbation to assess the impact on model performance
- Hyperparameter tuning to find the best performing parameters for each model
- Class imbalance can be addressed using random oversampling, undersampling, and weighted loss functions

S. Tipirneni and C. K. Reddy, "STraTS: Self-supervised transformer for time-series representation learning in healthcare," in *Proc. AAAI Conf. Artif. Intell.*, 2022.

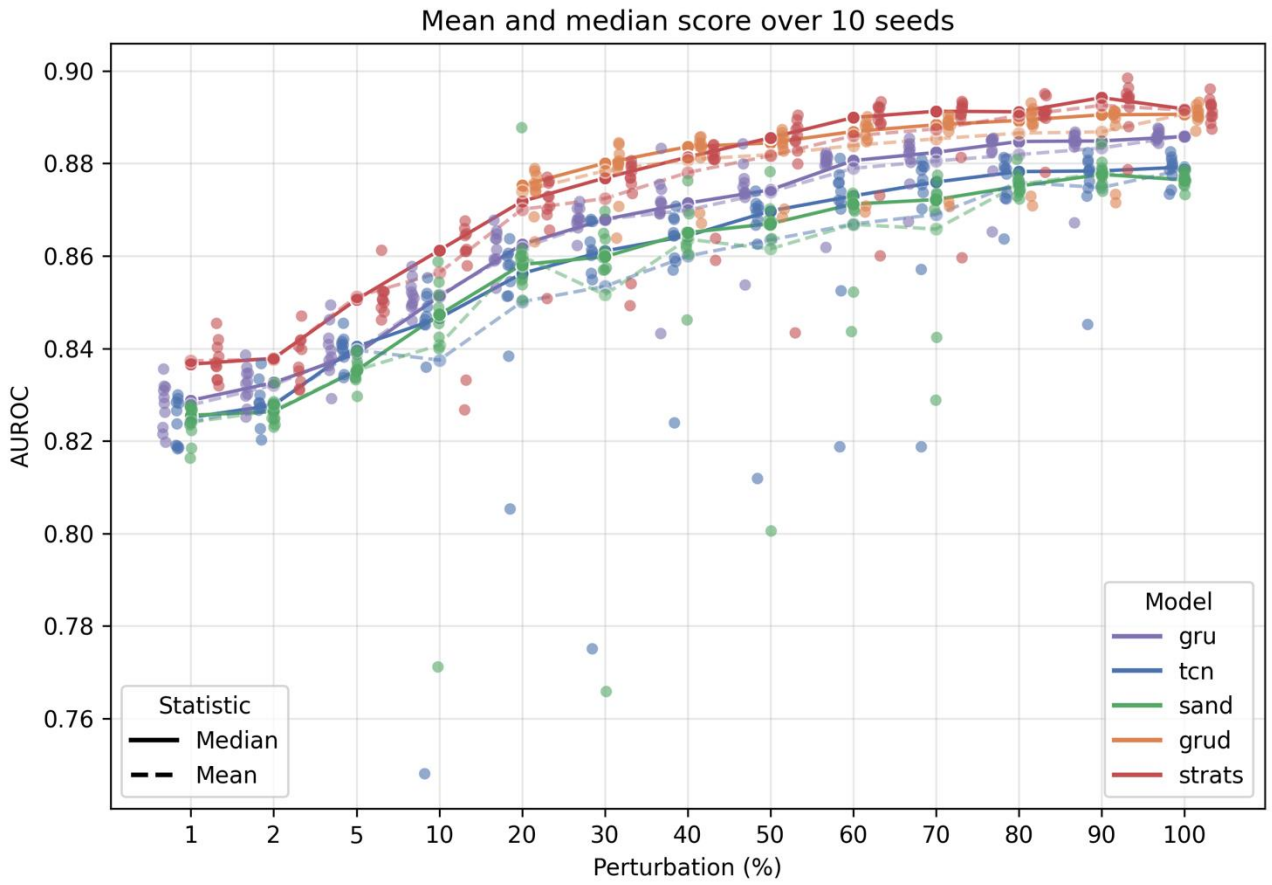
Thank You

Any Questions?

Additional Slides

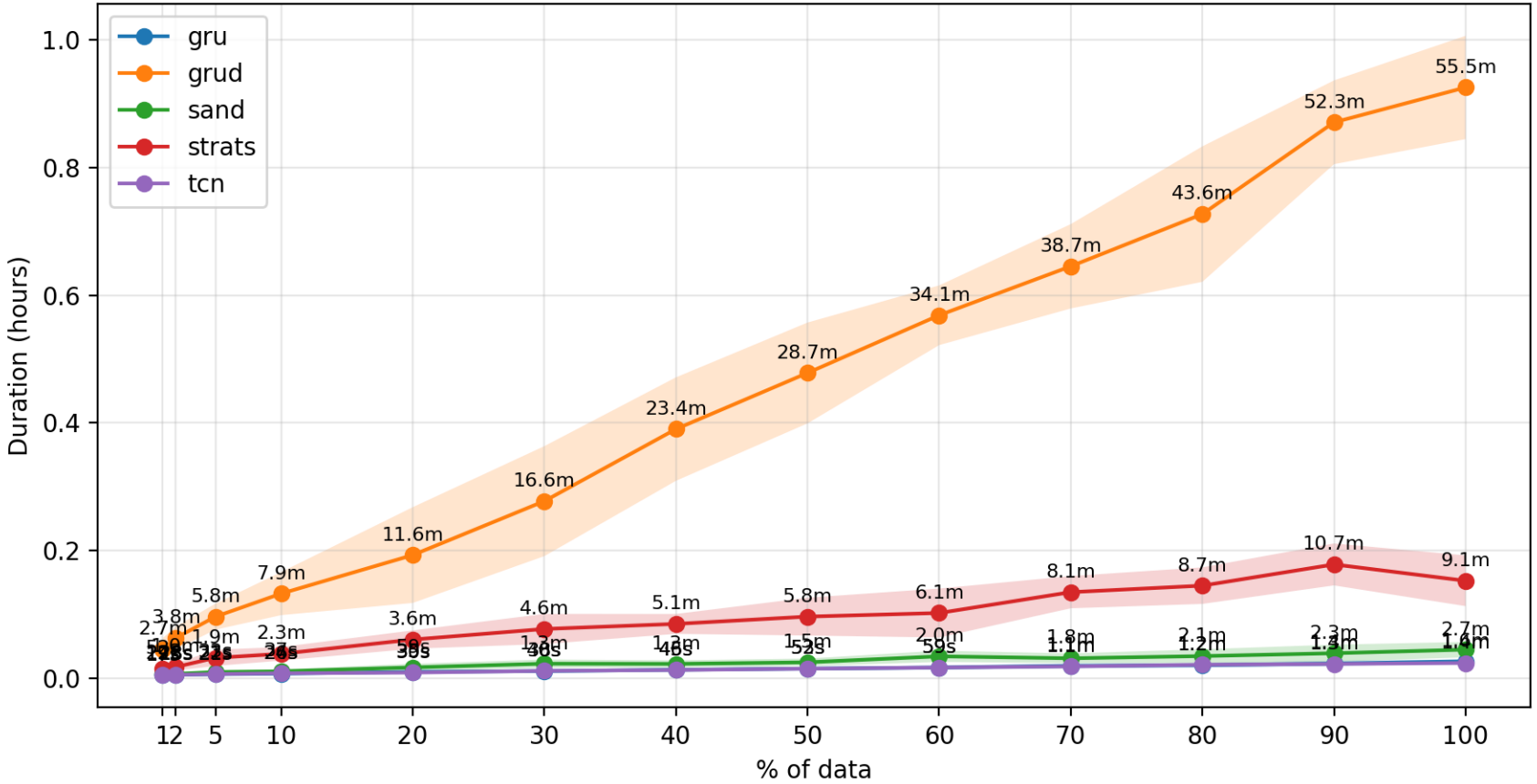
Additional

Sparsification (Mimic-III with GRU-D)



Duration

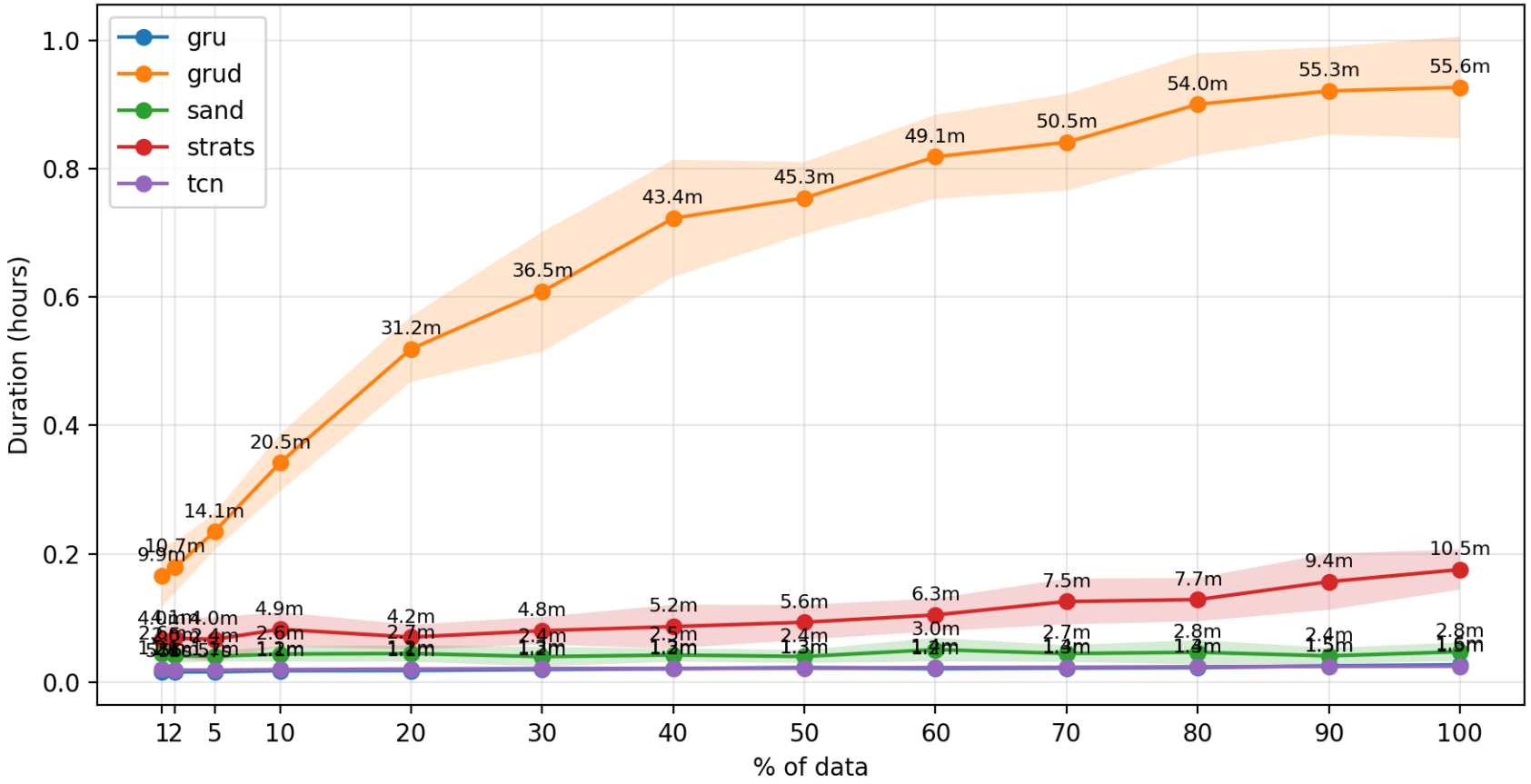
Subsample (Physionet-2012)



physionet_2012 | in_hospital_mortality | subsampled | device=cuda

Duration

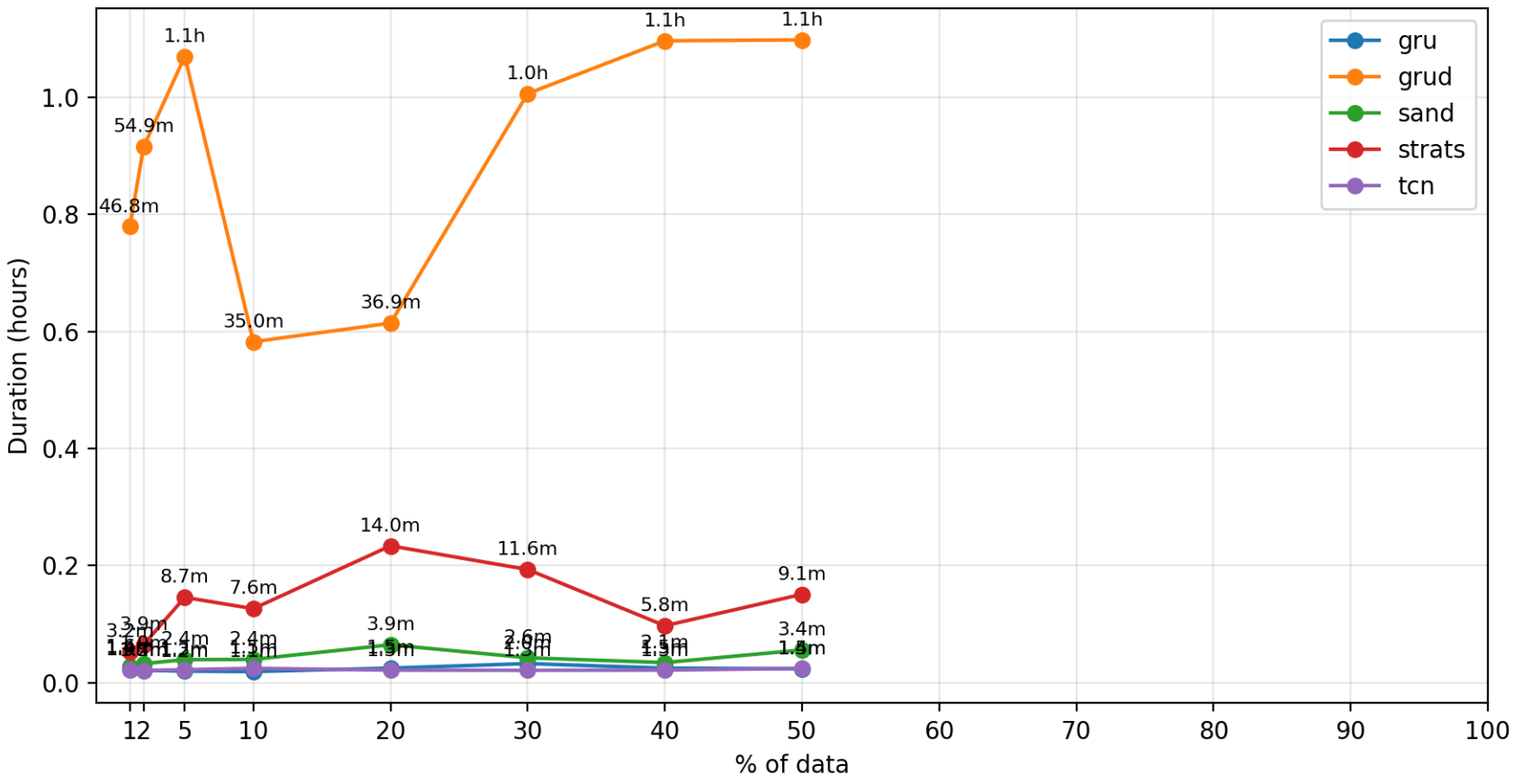
Sparsification (Physionet-2012)



physionet_2012 | in_hospital_mortality | sparsified-tsid-varid | device=cuda

Duration

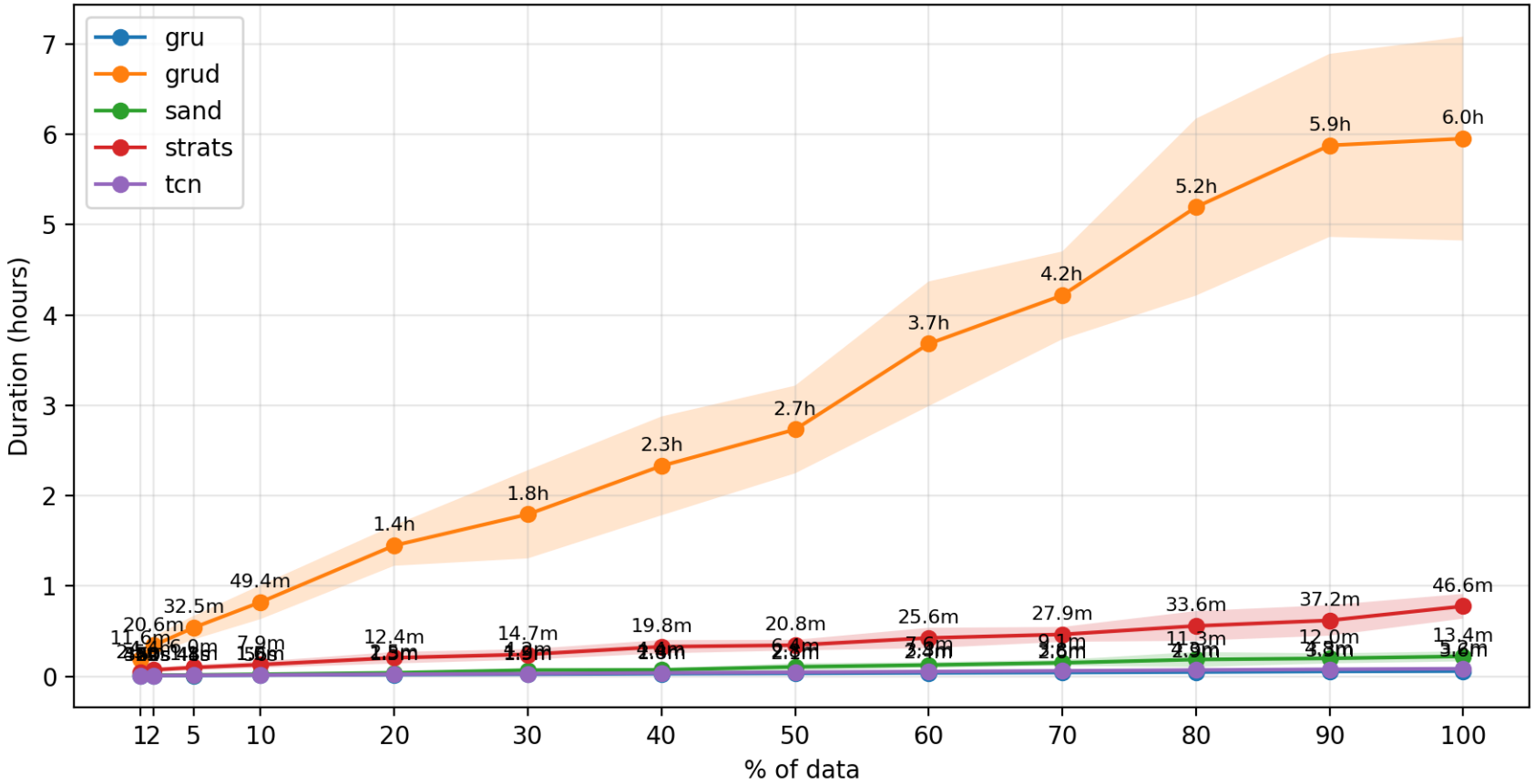
Imbalance (Physionet-2012)



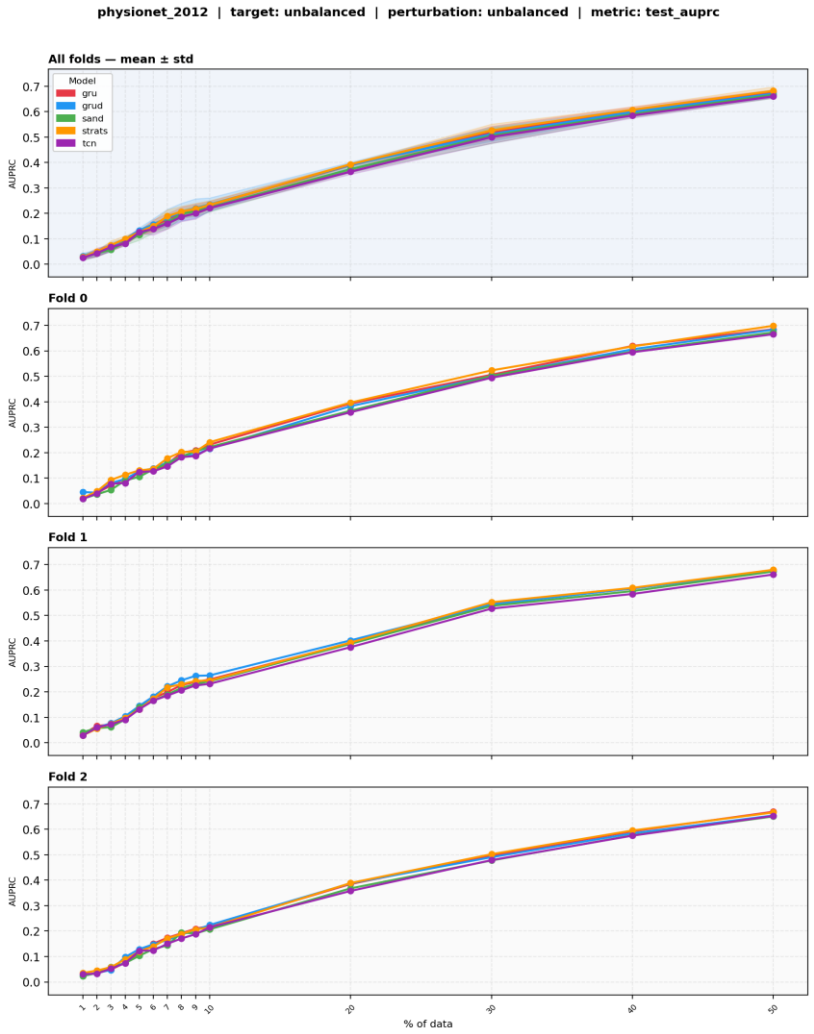
physionet_2012 | unbalanced | unbalanced | device=cuda

Duration

Subsample (Mimic-III)



mimic_iii | in_hospital_mortality | subsampled | device=cuda

Cross-Validation

Imbalanced (Physionet-2012)